

MicroBlend

Chen Notation ERD Explanation

Entity-Relationship Diagram Documentation for the
MicroBlend Point-of-Sale & Table Management System

Generated from ERD_Chen_System.mmd

MicroBlend — Chen Notation ERD Explanation

Overview

The Entity-Relationship Diagram (Chen notation) models the complete MicroBlend database — 26 entities and 35 relationships across 7 domain areas. Each entity is a rectangle, attributes are ovals, and relationships are diamonds with verbs and cardinality labels (1, N, M).

Domain 1: Users & Notifications

users

Central actor table supporting all account types: customers, waiters, kitchen/bar staff, cashiers, and admins. Guest accounts (is_guest) allow walk-in customers to order without registration. The soft-delete flag (is_deleted) preserves foreign key integrity when accounts are removed.

Relationships (14 outbound):

- `places` 1:N → orders
- `owns` 1:N → order_playlists
- `opens` 1:N → table_sessions (as the staff who opened)
- `sits at` 1:N → table_sessions (as the customer)
- `requests (scan)` 1:N → table_scan_requests
- `requests (staff)` 1:N → staff_page_requests
- `resolves` 1:N → staff_page_requests

- `receives` 1:N → notifications
- `performs` 1:N → audit_logs
- `submits` 1:N → feedback_entries
- `generates` 1:N → generated_reports
- `creates` 1:N → cost_simulations
- `triggers` 1:N → realtime_events
- `acts upon` 1:N → inventory_movements

notification

In-app alert records. Can target a specific user (recipient FK) or broadcast to all users of a given role (role_target string). Category enables client-side filtering.

debounce_record

Standalone idempotency guard with a unique constraint on (actor_key, action, object_key). No foreign keys to keep it fast — uses loose strings so it works for any actor type (user ID, device ID, system ID).

Domain 2: Menu & Ingredients

category

Groups menu items (e.g. Appetizers, Main Course, Desserts). Sort_order controls display order.

menu_item

Sellable items assigned to a preparation station (kitchen or bar). The is_available flag is toggled automatically when ingredient stock is insufficient.

menu_item_ingredient

M:N junction between menu_item and ingredient with a payload (quantity_required). The relationship itself carries the data — neither parent owns it alone.

ingredient

Raw materials tracked in inventory with a unit of measure (g/ml/pc) and a reorder_level threshold for low-stock alerts.

Domain 3: Inventory

inventory_batch

Records individual deliveries of an ingredient. Tracks quantity_added, quantity_remaining, unit_cost, and expiration_date for FIFO cost tracking and spoilage monitoring.

inventory_movement

Immutable log of every inventory transaction: restock, deduct, adjust, or delete. Links to the actor who performed it and optionally to the batch and order that caused the movement.

Domain 4: Orders

order

Core transaction record with three parallel status tracks (kitchen, bar, cashier) that advance independently. Denormalized fields (placed_by_name, placed_for_name) preserve context if related users are deleted. Channel identifies how the order entered (customer_account, guest, waiter_assisted, bulk, pos_sync).

order_item

Line items with denormalized copies of item_name, station, and unit_price at time of order. This snapshot ensures historical receipts remain accurate after menu changes.

order_status_log

Immutable audit trail for every status transition. Records who changed it, what it changed to, and optional JSON metadata.

feedback_entry

Customer feedback and problem reports. Links to both the submitting user and the order. Entry_type distinguishes feedback from reports; status tracks the handling lifecycle.

Domain 5: Order Playlists

order_playlist

Saved order template owned by a user. Unique on (owner_id, name) prevents duplicate named playlists.

order_playlist_item

Junction between playlist and menu_item with quantity. Unique on (playlist_id, menu_item_id).

Domain 6: Table Management

table

Physical tables with a display identifier, seating capacity, status (vacant/occupied/merged/blocked), zone for area grouping, and an auto-generated QR code value.

table_merge_group

Groups of merged tables (M:N with table via the implicit junction table_merge_group_tables). Allows flexible table combinations for large parties.

table_session

A seating session representing customer occupation of a table. Tracks which staff opened it, which customer is seated, how they arrived (source: qr/walk_in/waiter/manual), party size, and active duration.

table_scan_request

QR code scan intent from a customer. Records the table, the requesting user, the device used, and the moderation outcome (pending/approved/blocked).

staff_page_request

Customer request for staff assistance. Reason categorizes the need (cleanup/payment/accident). Links to both table and session for context, and tracks who resolved it.

Domain 7: Analytics, Integration & System

audit_log

Generic action log using a polymorphic FK pattern: target_type + target_id + target_label can reference any entity without formal FK constraints. Captures actor, action, role snapshot, and JSON metadata.

generated_report

Report generation request and result storage. Supports daily, weekly, monthly, annual, and custom date ranges. Report data is stored as JSON payload.

cost_simulation

What-if cost analysis scenarios. Users input assumptions (ingredient prices, quantities, waste) as JSON; the system computes projected costs and margins stored as results JSON.

external_system

Registered integration partners (POS, mobile, kiosk). Type and active flag control routing and enable/disable sync without removing configuration.

sync_event

Outbox-pattern event log for external system sync. Events are written during the main request and consumed asynchronously. The idempotency_key prevents duplicate processing.

realtime_event

SSE (Server-Sent Events) data store. Events are created by domain services and consumed by clients via the polling-based SSE endpoint at /api/realtime/stream/. Targeting by role or user restricts which clients receive each event.

Key Notation Conventions

Element	Chen Symbol	Meaning
Entity	Rectangle	A table in the database
Attribute	Oval	A column/field of the table
Relationship	Diamond	A foreign key or link between tables
1	On edge near entity	"One" side of the relationship
N, M	On edge near entity	"Many" side of the relationship
(wk)	Entity label	Weak/associative entity (junction table)
(m2m)	Relationship label	Implicit many-to-many relationship

Summary Statistics

Metric	Count
Entities	26
Relationships	35
User outbound relationships	14
Weak / associative entities	6